Table 17.1 Launch template parameters

Name	Description	Possible values
`ImageId`	Image from which to start a virtual machine	ID of an Amazon Machine Image (AMI)
`InstanceType`	Size for new virtual machines	Instance type (such as `t2.micro`)
`UserData`	User data for the virtual machine used to execute a script during bootstrapping	BASE64-encoded string
`NetworkInterfaces`	Configures the network interfaces of the virtual machine. Most importantly, this parameter allows you to attach a public IP address to the instance.	List of network interface configurations
`IamInstanceProfile`	Attaches an IAM instance profile linked to an IAM role	Name or Amazon Resource Name (ARN, an ID) of an IAM instance profile

After you create a launch template, you can create an Auto Scaling group that references it. The Auto Scaling group defines the maximum, minimum, and desired number of virtual machines. *Desired* means this number of EC2 instances should be running. If the current number of EC2 instances is below the desired number, the Auto Scaling group will add EC2 instances. If the current number of EC2 instances is above the desired number, EC2 instances will be terminated. The desired capacity can be changed automatically based on load or a schedule, or manually. *Minimum* and *maximum* are the lower and upper limits for the number of virtual machines within the Auto Scaling group.

The Auto Scaling group also monitors whether EC2 instances are healthy and replaces broken instances. Table 17.2 shows the most important parameters for an Auto Scaling group.